

Evan Edelstein
 EN.605.645.82.SP26
 Module 12 – self check

- 1) The number of features in X is 2. Using the Euclidian distance formula, we can calculate the distance from example 6 to the other examples:

$$D_1 = \sqrt{|0.23 - 0.39|^2 + |0.81 - 0.63|^2} = 0.24$$

$$D_2 = \sqrt{|0.42 - 0.39|^2 + |0.78 - 0.63|^2} = 0.15$$

$$D_3 = \sqrt{|0.64 - 0.39|^2 + |0.23 - 0.63|^2} = 0.47$$

$$D_4 = \sqrt{|0.87 - 0.39|^2 + |0.19 - 0.63|^2} = 0.65$$

$$D_5 = \sqrt{|0.76 - 0.39|^2 + |0.43 - 0.63|^2} = 0.42$$

- a) The 3 closest neighbors are the three lowest distance examples, 2, 1 and 5.
 b) The predicted value for 6 would be the average of the y value for example 2, 1 and 5, which is $\frac{0.33+0.18+0.32}{3} = 0.28$.
 c) The square root is not required to compare distances, since we only care about which examples are closest to the query example, not the distance value itself.

2) Metrics:

$$a. \text{ Accuracy} = \frac{TP+TN}{n} = \frac{329+357}{329+35+87+357} = \frac{686}{808} = 0.85$$

$$b. \text{ error} = \frac{FP+FN}{n} = \frac{35+87}{329+35+87+357} = \frac{122}{808} = 0.15$$

$$c. \text{ precision} = \frac{TP}{TP+FP} = \frac{329}{329+35} = \frac{329}{364} = 0.90$$

$$d. \text{ recall} = \frac{TP}{TP+FN} = \frac{329}{329+87} = \frac{329}{416} = 0.79$$

3)

y	y_hat	y - y_hat	(y-yhat)^2	y-mean	(y-mean)^2
3.78	3.32	0.46	0.2116	0.246	0.060516
4.82	5.21	-0.39	0.1521	1.286	1.653796
2.83	2.97	-0.14	0.0196	-0.704	0.495616
2.76	2.37	0.39	0.1521	-0.774	0.599076
3.48	3.29	0.19	0.0361	-0.054	0.002916
MSE	0.1143				
mean	3.534				
Mean MSE	0.562384				

Population variance is the same as mean MSE

- 4) The first curve shows a high bias case where more data will not help, while the second curve shows a high variance case where more data may help.

